

Generative Models in Finance — Coursework

Spring 2026 · Christopher Salvi · Imperial College London

Deadline: 1 April 2026 · Groups of 2 · Weight: 100%

Deliverables. Submit a single .zip file to c.salvi@imperial.ac.uk by the deadline containing:

- One report (PDF) covering Parts I and III.
- A Jupyter notebook (.ipynb) and any auxiliary files for Part II.

Part I: Mathematics — From Denoising to Flow Matching

The following papers are **required reading** for this part. The problems below guide you through their key results.

- [H] A. Hyvärinen. *Estimation of Non-Normalized Statistical Models by Score Matching*. JMLR, 2005.
- [V] P. Vincent. *A Connection Between Score Matching and Denoising Autoencoders*. Neural Computation, 2011.
- [SE] Y. Song and S. Ermon. *Generative Modeling by Estimating Gradients of the Data Distribution*. NeurIPS, 2019.
- [Ho] J. Ho, A. Jain, and P. Abbeel. *Denoising Diffusion Probabilistic Models*. NeurIPS, 2020.
- [S+] Y. Song et al. *Score-Based Generative Modeling through SDEs*. ICLR, 2021.
- [L+] Y. Lipman et al. *Flow Matching for Generative Modeling*. ICLR, 2023.
- [A+] M.S. Albergo, N.M. Boffi, and E. Vanden-Eijnden. *Stochastic Interpolants: A Unifying Framework for Flows and Diffusions*. arXiv:2303.08797, 2023.

Throughout, let $z \sim p_{\text{data}}$ on $\mathbb{R}^d$, $\epsilon \sim \mathcal{N}(0, I_d)$ independent of z , and $x = \alpha_t z + \beta_t \epsilon$ for noise schedulers α_t, β_t with $\alpha_0 = \beta_1 = 0$, $\alpha_1 = \beta_0 = 1$. The conditional distribution is $p_t(x | z) = \mathcal{N}(\alpha_t z, \beta_t^2 I_d)$.

Problem 1: Implicit score matching (Hyvärinen, 2005). The score $\nabla \log p(x)$ of a distribution p is unknown, yet we can still set up a tractable objective to learn it. Here p denotes a generic density on $\mathbb{R}^d$ (not necessarily p_t).

- (a) Let $s_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a parametric model. Write down the **explicit score matching** objective:

$$J_{\text{ESM}}(\theta) = \frac{1}{2} \mathbb{E}_{x \sim p} [\|s_\theta(x) - \nabla \log p(x)\|^2].$$

Expand the square and identify the term that depends on θ but involves the unknown $\nabla \log p$.

- (b) Using integration by parts, show that for each component i :

$$\mathbb{E}_{x \sim p} [s_{\theta,i}(x) \partial_{x_i} \log p(x)] = -\mathbb{E}_{x \sim p} [\partial_{x_i} s_{\theta,i}(x)],$$

provided $p(x) s_{\theta,i}(x) \rightarrow 0$ as $\|x\| \rightarrow \infty$.

(c) Conclude that, up to a constant independent of θ :

$$J_{\text{ESM}}(\theta) = \mathbb{E}_{x \sim p} \left[\sum_{i=1}^d \partial_{x_i} s_{\theta,i}(x) + \frac{1}{2} \|s_{\theta}(x)\|^2 \right] + C.$$

This is the **implicit score matching** (ISM) objective: it only requires samples from p , not $\nabla \log p$ itself.

Problem 2: Tweedie's formula. *The posterior mean $\mathbb{E}[z \mid x]$ can be expressed in terms of the marginal score $\nabla \log p_t(x)$.*

(a) Show that $\nabla_x p_t(x \mid z) = -\frac{x - \alpha_t z}{\beta_t^2} p_t(x \mid z)$.

(b) Using $p_t(x) = \int p_t(x \mid z) p_{\text{data}}(z) dz$, show that

$$\nabla \log p_t(x) = -\frac{x - \alpha_t \mathbb{E}[z \mid x]}{\beta_t^2}.$$

(c) Deduce **Tweedie's formula**:

$$\mathbb{E}[z \mid x] = \frac{1}{\alpha_t} (x + \beta_t^2 \nabla \log p_t(x)).$$

Problem 3: Denoising score matching. *We prove that regressing a network $s_t^\theta(x)$ against the conditional score $\nabla \log p_t(x \mid z)$ is equivalent to matching the (intractable) marginal score $\nabla \log p_t(x)$.*

(a) Write down the **marginal** and **conditional** score matching losses:

$$\mathcal{L}_{\text{SM}}(\theta) = \mathbb{E}_{t, x \sim p_t} [\|s_t^\theta(x) - \nabla \log p_t(x)\|^2], \quad \mathcal{L}_{\text{CSM}}(\theta) = \mathbb{E}_{t, z, x \sim p_t(\cdot \mid z)} [\|s_t^\theta(x) - \nabla \log p_t(x \mid z)\|^2].$$

Expand $\mathcal{L}_{\text{SM}}$ as $\mathbb{E}[\|s_t^\theta\|^2] - 2\mathbb{E}[(s_t^\theta)^T \nabla \log p_t] + C_1$, where C_1 is independent of θ .

(b) Show that the cross term satisfies

$$\mathbb{E}_{t, x \sim p_t} [(s_t^\theta(x))^T \nabla \log p_t(x)] = \mathbb{E}_{t, z, x \sim p_t(\cdot \mid z)} [(s_t^\theta(x))^T \nabla \log p_t(x \mid z)].$$

(c) Conclude that $\mathcal{L}_{\text{SM}}(\theta) = \mathcal{L}_{\text{CSM}}(\theta) + C$ for a constant C independent of θ .

Problem 4: From score matching to DDPM. *We show that the denoising diffusion (DDPM) ϵ -prediction loss is a reparametrisation of conditional score matching.*

(a) Using Problem 2(a), show that $\nabla \log p_t(x \mid z) = -\epsilon/\beta_t$ when $x = \alpha_t z + \beta_t \epsilon$.

(b) Define the **noise predictor** $\epsilon_t^\theta(x) := -\beta_t s_t^\theta(x)$. Substitute into $\mathcal{L}_{\text{CSM}}$ to show

$$\mathcal{L}_{\text{CSM}}(\theta) = \frac{1}{\beta_t^2} \mathbb{E}_{t, z, \epsilon} [\|\epsilon_t^\theta(\alpha_t z + \beta_t \epsilon) - \epsilon\|^2].$$

(c) Explain why minimising $\mathcal{L}_{\text{CSM}}$ is equivalent to minimising the **DDPM loss** $\mathcal{L}_{\text{DDPM}}(\theta) = \mathbb{E}_{t, z, \epsilon} [\|\epsilon_t^\theta(\alpha_t z + \beta_t \epsilon) - \epsilon\|^2]$. What is the role of the $1/\beta_t^2$ weighting?

Problem 5: Equivalence with flow matching. *We show that, for Gaussian paths, a trained score network can be converted into a flow matching vector field, and vice versa.*

- (a) Recall the conditional vector field $u_t^{\text{target}}(x | z) = (\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t} \alpha_t)z + \frac{\dot{\beta}_t}{\beta_t}x$ from the lectures. Show that it can be rewritten as

$$u_t^{\text{target}}(x | z) = \frac{\dot{\alpha}_t}{\alpha_t}x + \left(\frac{\dot{\alpha}_t \beta_t^2}{\alpha_t} - \dot{\beta}_t \beta_t \right) \nabla \log p_t(x | z).$$

- (b) Using the identity from Problem 3(b) (replacing scores by vector fields via the same marginalisation structure), argue that the same conversion holds at the marginal level:

$$u_t^{\text{target}}(x) = \frac{\dot{\alpha}_t}{\alpha_t}x + \left(\frac{\dot{\alpha}_t \beta_t^2}{\alpha_t} - \dot{\beta}_t \beta_t \right) \nabla \log p_t(x).$$

- (c) Specialise to the CondOT path ($\alpha_t = t$, $\beta_t = 1 - t$). Write out $u_t^{\text{target}}(x)$ explicitly in terms of x and $\nabla \log p_t(x)$, and verify that $u_t^{\text{target}}(x | z) = z - \epsilon$ when $x = tz + (1 - t)\epsilon$.

Problem 6: The probability flow ODE. *We derive a deterministic ODE whose trajectories have the same marginal distributions as a given SDE — this is the probability flow ODE (Song et al., 2021).*

- (a) Recall the **Fokker–Planck equation** from the lectures: the SDE $dX_t = u_t(X_t)dt + \sigma_t dW_t$ with $X_0 \sim p_0$ satisfies $X_t \sim p_t$ if and only if

$$\partial_t p_t = -\text{div}(p_t u_t) + \frac{\sigma_t^2}{2} \Delta p_t.$$

Using $\Delta p_t = \text{div}(\nabla p_t) = \text{div}(p_t \nabla \log p_t)$, rewrite the Fokker–Planck equation as a *continuity equation*:

$$\partial_t p_t = -\text{div}(p_t \tilde{u}_t), \quad \tilde{u}_t(x) = ?$$

Identify $\tilde{u}_t$ explicitly in terms of u_t , σ_t , and $\nabla \log p_t$.

- (b) The continuity equation $\partial_t p_t = -\text{div}(p_t \tilde{u}_t)$ corresponds to the ODE $dX_t = \tilde{u}_t(X_t)dt$. Deduce the **probability flow ODE**:

$$dX_t = \left[u_t(X_t) - \frac{\sigma_t^2}{2} \nabla \log p_t(X_t) \right] dt.$$

Explain in one sentence why this ODE and the original SDE produce the same marginal distributions p_t despite having different individual trajectories.

- (c) Using the conversion formula from Problem 5(b), show that for Gaussian paths the probability flow ODE can be written purely in terms of the vector field u_t^{target} :

$$dX_t = u_t^{\text{target}}(X_t)dt.$$

Part II: Coding — Flow Matching on 2D Distributions

Implement conditional flow matching from scratch in PyTorch. Train a neural network vector field u_t^θ on a synthetic 2D dataset and use it to generate new samples. Submit a Jupyter notebook (.ipynb) with code, outputs, and brief commentary. All experiments should run on CPU in under 10 minutes.

Task 1: Data and probability paths.

- (a) Choose a non-trivial 2D target distribution p_{data} (e.g. two moons, concentric circles, a mixture of 8 Gaussians arranged in a ring, or a Swiss roll). Write a function `sample_data(n)` that returns n samples.
- (b) Implement the CondOT probability path: given a data sample z and noise $\epsilon \sim \mathcal{N}(0, I_2)$, compute $x_t = tz + (1 - t)\epsilon$ and the target vector field $u_t^{\text{target}}(x_t | z) = z - \epsilon$. Visualise samples from $p_t(\cdot | z)$ for a fixed z at times $t \in \{0, 0.25, 0.5, 0.75, 1\}$.

Task 2: Training.

- (a) Define a neural network $u_t^\theta(x)$ that takes as input $(x, t) \in \mathbb{R}^2 \times [0, 1]$ and outputs a vector in $\mathbb{R}^2$. Use an MLP with at least 3 hidden layers (suggested width: 128). Time t should be concatenated to x as an input feature.
- (b) Implement Algorithm 3 from the lectures (flow matching training with CondOT path). Train for a sufficient number of iterations (suggested: 10 000–20 000 steps, batch size 256, learning rate 10^{-3} with Adam). Plot the training loss curve.

Task 3: Sampling and evaluation.

- (a) Implement Algorithm 1 from the lectures (Euler method sampling). Generate 2 000 samples from the trained model using $n = 100$ Euler steps. Plot the generated samples alongside true samples from p_{data} .
- (b) Implement Heun’s method (second-order) for sampling. Compare the quality of generated samples using Euler vs. Heun at $n \in \{5, 10, 25, 50, 100\}$ steps. Summarise your findings in a table or figure.
- (c) Visualise the learned vector field: plot $u_t^\theta(x)$ as a quiver plot on a grid at times $t \in \{0, 0.25, 0.5, 0.75, 1\}$. Overlay a few ODE trajectories from $t = 0$ to $t = 1$.

Task 4: Extending to diffusion (bonus).

- (a) Using your trained vector field u_t^θ and the conversion formula from Problem 5, derive a score network s_t^θ for the CondOT path. Implement Euler–Maruyama sampling (Algorithm 2 from the lectures) with $\sigma_t = 0.1$. Compare the generated samples with the ODE samples from Task 3.

Part III: Literature Review — Discrete Diffusion and Flow Matching for Text

The continuous-state diffusion and flow matching framework studied in Parts I–II operates on $\mathbb{R}^d$. A major open question is how to extend these ideas to **discrete state spaces** (e.g. token sequences), where the data lives on a finite set rather than in Euclidean space. This is the domain of autoregressive LLMs (Weeks 1–3 of this course). Recently, discrete diffusion and flow matching models have emerged as a potential alternative to autoregressive generation for text — notably through the startup Inception Labs (co-founded by S. Ermon), whose diffusion-based language model *Mercury* demonstrates competitive quality with significantly faster inference than autoregressive models.

Write a critical review of at most 5 pages (excluding references) on the topic of discrete diffusion and flow matching for text generation. Your review should:

- *Summarise the main contributions of this line of work and situate it within the landscape of generative models for discrete data.*
- *Identify the key mathematical idea(s): what replaces the score function, the SDE, and/or the probability path when the state space is discrete?*
- *Critically compare with autoregressive LLMs: what are the architectural, training, and inference trade-offs? Under what conditions might discrete diffusion/flow matching models outperform or complement autoregressive generation?*

The following references provide a good starting point for this review:

1. J. Austin, D.D. Johnson, J. Ho, D. Tarlow, and R. van den Berg. *Structured Denoising Diffusion Models in Discrete State-Spaces* (D3PM). NeurIPS, 2021.
2. A. Lou, C. Meng, and S. Ermon. *Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution* (SEDD). ICML, 2024.
3. A. Campbell, J. Benton, V. De Bortoli, T. Rainforth, G. Deligiannidis, and A. Doucet. *A Continuous Time Framework for Discrete Denoising Models*. NeurIPS, 2022.
4. J. Shi, K. Han, Z. Wang, A. Doucet, and M. Titsias. *Simplified and Generalized Masked Diffusion for Discrete Data*. NeurIPS, 2024.
5. I. Gat, T. Remez, N. Shaul, F. Kreuk, R.T.Q. Chen, G. Synnaeve, Y. Adi, and Y. Lipman. *Discrete Flow Matching*. NeurIPS, 2024.
6. S.S. Sahoo, M. Arriola, Y. Schiff, A. Gokaslan, E. Marroquin, J.T. Chiu, A. Rush, and V. Kuleshov. *Simple and Effective Masked Diffusion Language Models*. NeurIPS, 2024.